

Yixuan Huang

✉ yixuan.huang@princeton.edu ☎ +1-858-250-9569 🔗 <https://yixuanhuang98.github.io/>

Research Experience

Postdoctoral Research Associate with Prof. Tom Silver <i>Princeton University</i>	<i>Princeton, NJ</i> Aug 2025 – Present
Visiting Student Researcher with Prof. Jeannette Bohg <i>Stanford University</i>	<i>Stanford, CA</i> Jan 2024 – Aug 2024
Graduate Research Assistant with Prof. Tucker Hermans <i>University of Utah</i>	<i>Salt Lake City, UT</i> May 2021 – Aug 2025
Undergraduate Research Assistant with Prof. Sicun Gao <i>University of California, San Diego</i>	<i>La Jolla, CA</i> Aug 2019 – Aug 2020

Education

University of Utah, Salt Lake City, UT <i>Ph.D. in Computing: Robotics</i> ◦ GPA: 3.98/4.0 ◦ Advisor: Prof. Tucker Hermans	Aug 2020 - Aug 2025
Northeastern University, Shenyang, Liaoning (China) <i>B.E. in Computer Science and Technology (<u>top</u> student in the department)</i> ◦ GPA: 93.2/100, Rank: 1/278	Sep 2016 - Jun 2020
University of California, San Diego, La Jolla, CA <i>Exchange student</i> ◦ GPA: 3.91/4	Sep 2018 - Jun 2019

Publications

Ordered by recency. *Equal contribution. †Student mentored by me.

Under Review

- [1] **Learning By and For Task and Motion Planning: A Survey**
In Submission, 2026
Yixuan Huang*, Benned Hedegaard*, Naman Shah, Siddharth Srivastava, George Konidaris, and Tom Silver
[Paper]
- [2] **TANDEM: Task and Motion Planning with As-Needed Demonstrations for Efficient Vision-Language-Action Model Fine-tuning**
In Submission, 2026
Samrat Sahoo[†], Liang Ji[†], Tom Silver, and **Yixuan Huang**
[Project Website]
- [3] **DATAFARM: Distribution-Aligned Task and Motion Planning for Fine-Tuning Vision-Language-Action Models**
In Submission, 2026
Samrat Sahoo[†], **Yixuan Huang**, and Tom Silver
[Project Website] [Paper]
- [4] **Chaining Visuomotor Skills for Constrained Manipulation Through Perceptual Manipulation**
In Submission, 2026
Hao Wang[†], **Yixuan Huang**, Somil Bansal, and Tom Silver
[Project Website]
- [5] **SPECTRE: Skeleton Priority Estimation Conditioned on TAMP Refinement Evidence**

In Submission, 2026

Jiacheng Xu[†], **Yixuan Huang**, and Tom Silver

[Project Website]

[6] **Coding Agents for Generalized Task and Motion Planning Problems**

In Submission, 2026

Matteo Merler, Bowen Li, Josh Roy, Yichao Liang, Qianwei Wang, **Yixuan Huang**, and Tom Silver

[Project Website]

[7] **Skeleton-Conditioned Base Placement Evaluation for Long-Horizon Mobile Manipulation**

In Submission, 2026

Qihang Li[†], **Yixuan Huang**, Tom Silver, and Peng Gao

Journal Publications

[1] **Latent Space Planning for Multi-Object Manipulation with Environment-Aware Relational Classifiers**

IEEE Transactions on Robotics (T-RO), 2024

Yixuan Huang, Nichols Crawford Taylor[†], Adam Conkey, Weiyu Liu, and Tucker Hermans

[Project Website] [Paper]

Conference Publications

[1] **KinDER: A Physical Reasoning Benchmark for Robot Learning and Planning**

Robotics: Science and Systems (RSS), 2026

Yixuan Huang^{*}, Bowen Li^{*}, Vaibhav Saxena^{*}, Yichao Liang, Utkarsh Aashu Mishra, Liang Ji[†], Lihan Zha, Jimmy Wu, Nishanth Kumar, Sebastian Scherer, Danfei Xu, and Tom Silver

[Project Website] [Paper]

[2] **RoboVista: Evaluating Vision Language Models for Diverse Robot Applications**

Robotics: Science and Systems (RSS), 2026

Shuangyu Xie^{*}, Kaiyuan Chen^{*}, Ziyang Chen, Simeon Adebola, **Yixuan Huang**, Zehan Ma, Tianshuang Qiu, Wentao Yuan, Dhruv Shah, Pannag R Sanketi, and Ken Goldberg

[Project Website] [Paper]

[3] **LAP: Language-Action Pre-training Enables Zero-Shot Cross-Embodiment Transfer**

Robotics: Science and Systems (RSS), 2026

Lihan Zha, Asher James Hancock^{*}, Mingtong Zhang^{*}, Tenny Yin, **Yixuan Huang**, Dhruv Shah, Allen Z. Ren, and Anirudha Majumdar

[Project Website] [Paper]

[4] **Hierarchical DLO Routing with Reinforcement Learning and Multimodal In-Context VLMs**

IEEE International Conference on Robotics and Automation (ICRA), 2026

Best Paper Award on Robot Learning – Finalist

Mingen Li, Houjian Yu, **Yixuan Huang**, Youngjin Hong, and Changhyun Choi

[Project Website] [Paper]

[5] **Fail2Progress: Learning from Real-World Robot Failures with Stein Variational Inference**

Conference on Robot Learning (CoRL), 2025

Yixuan Huang, Novella Alvina[†], Mohanraj Devendran Shanthi, and Tucker Hermans

[Project Website] [Paper]

[6] **Points2Plans: From Point Clouds to Long-Horizon Plans with Composable Relational Dynamics**

IEEE International Conference on Robotics and Automation (ICRA), 2025

Yixuan Huang, Christopher Agia, Jimmy Wu, Tucker Hermans, and Jeannette Bohg

[Project Website] [Paper]

[7] **Out of Sight, Still in Mind: Reasoning and Planning about Unobserved Objects with Video Tracking Enabled Memory Models**

IEEE International Conference on Robotics and Automation (ICRA), 2024

Yixuan Huang, Jialin Yuan, Chanh Kim, Pupul Pradhan[†], Bryan Chen, Li Fuxin, and Tucker Hermans

[Project Website] [Paper]

[8] **Planning for Multi-Object Manipulation with Graph Neural Network Relational Classifiers**

IEEE International Conference on Robotics and Automation (ICRA), 2023
Yixuan Huang, Adam Conkey, and Tucker Hermans
[Project Website] [Paper]

- [9] **Toward Learning Context-Dependent Tasks from Demonstration for Tendon-Driven Surgical Robots**
International Symposium on Medical Robotics (ISMR), 2021
Best Paper Award – Finalist & Best Student Paper Award – Finalist
Yixuan Huang, Michael Bentley, Tucker Hermans, and Alan Kuntz
[Paper]

Workshop Papers

- [1] **Learning Multimodal Probabilistic Models of Manipulation Skill Effects**
CoRL 2025 Workshop on Learning to Simulate Robot Worlds
Siyeon Kim, Mohanraj Devendran Shanthi, **Yixuan Huang**, and Tucker Hermans
- [2] **Fail2Progress: Learning from Failures with Stein Variational Inference for Robot Manipulation**
RSS 2025 Workshop on Out-of-Distribution Generalization in Robotics: Towards Reliable Learning-Based Autonomy
Best Paper Award – Finalist
Yixuan Huang, Novella Alvina[†], Mohanraj Devendran Shanthi, and Tucker Hermans
- [3] **Points2Plans: From Point Clouds to Long-Horizon Plans with Composable Relational Dynamics**
CoRL 2024 Workshop on Learning Effective Abstractions for Planning (LEAP)
Oral Presentation
Yixuan Huang, Christopher Agia, Jimmy Wu, Tucker Hermans, and Jeannette Bohg
- [4] **Latent Space Planning for Unobserved Objects with Environment-Aware Relational Classifiers**
IROS 2023 Workshop on Causality for Robotics
Yixuan Huang, Jialin Yuan, Weiyu Liu, Chanh Kim, Li Fuxin, and Tucker Hermans
- [5] **Reasoning and Planning about Unobserved Objects with Memory Models**
CoRL 2023 Workshop on Neural Representation Learning for Robot Manipulation
Yixuan Huang

Honors and Awards

2026 ICRA Best Paper Award on Robot Learning – Finalist	May 2026
Rising Star at the Princeton Robotics Inaugural Symposium	April 2026
2025 ICRA Doctoral Consortium	May 2025
2021 ISMR Best Paper Award – Finalist	Nov 2021
2021 ISMR Best Student Paper Award – Finalist	Nov 2021
2021 ISMR NSF Travel Award	Oct 2021
University of Utah School of Computing Department Fellowship	Aug 2020
National Scholarship (top 1%)	Nov 2017 & 2018
Northeastern University Excellent Student (top 1%)	Dec 2017 & 2018
Second Place in National Mathematical Modeling Competition in China	Oct 2017
First Place in Provincial Mathematical Modeling Competition	Oct.2017

Invited Talks

- Sep 2026, "Toward Structured and Adaptive Robot Physical Reasoning", Host: Rahul Mangharam, Brown Robotics Seminar, Providence, RI.
- June 2026, "Toward Structured and Adaptive Robot Physical Reasoning", Host: Rahul Mangharam, University of Pennsylvania, Philadelphia, PA.
- Mar 2026, "Toward Structured and Adaptive Robot Physical Reasoning", Host: Felix Heide, Princeton University, Princeton, NJ.
- Sep 2025, "Planning for Sequential Multi-Object Robot Manipulation with Learned Relational Dynamics Models", Princeton Postdoctoral Council Seminar Series, Princeton, NJ.

July 2025, "Planning for Sequential Multi-Object Robot Manipulation with Learned Relational Dynamics Models", Oregon State University, Host: Li Fuxin, Corvallis, OR.

May 2025, "Planning for Sequential Multi-Object Robot Manipulation with Learned Relational Dynamics Models", University of Texas at Austin, Host: Roberto Martín-Martín, Austin, TX.

Apr 2025, "From Sensor Data to Long-Horizon Plans with Spatial-Temporal Reasoning", University of Utah Robotics Seminar, Host: Haohan Zhang, Salt Lake City, UT.

Jan 2025, "From Sensor Data to Long-Horizon Plans with Spatial-Temporal Reasoning", University of Utah Graduate Research Symposium, Host: Pratik Soni, Salt Lake City, UT.

Feb 2024, "Reasoning and Planning for Unobserved Objects with Environment-Aware Relational Classifiers and Memory Models", University of Illinois Urbana-Champaign, Host: Katie Driggs-Campbell, Champaign, IL.

Nov 2021, "Toward Learning Context-Dependent Tasks from Demonstration for Tendon-Driven Surgical Robots", University of Utah Robotics Seminar, Host: Jake J Abbott, Salt Lake City, UT.

Teaching

ECE 531/COS 531: Robot Planning Meets Machine Learning <i>Guest Lecture about Learning Models for Robot Planning</i>	<i>Princeton University</i> Oct 2025
CS 6300 Artificial Intelligence <i>Guest Lecture about Reinforcement Learning</i>	<i>University of Utah</i> Nov 2024
CS 4300 Artificial Intelligence <i>Guest Lecture about Imitation Learning</i>	<i>University of Utah</i> April 2022
CS 4300 Artificial Intelligence <i>Teaching Assistant</i>	<i>University of Utah</i> Jan 2022 – May 2022
CS 4300 Artificial Intelligence <i>Teaching Assistant</i>	<i>University of Utah</i> Aug 2022 – Dec 2022

Mentoring

Mentees who co-authored the publications listed above are marked with [†].

Samrat Sahoo [†] (Princeton Visitor → Stanford Master)	Summer 2026
Hao Wang [†] (Princeton Visitor → Stanford PhD)	Summer 2026
Liang Ji [†] (Princeton Visitor → Stanford Master)	Fall 2025 - Summer 2026
Jiacheng (Joseph) Xu [†] (Princeton Master)	Fall 2025 - Present
Qihang Li [†] (NCSU PhD)	Spring 2026 - Present
Ian Henriques (Princeton Master → MIT PhD)	Fall 2025 - Spring 2026
Tom Wang (Princeton Undergrad)	Fall 2025
Akash Bhowmick (Princeton Undergrad)	Fall 2025
Nichols Crawford Taylor [†] (Utah Undergrad → Northeastern PhD)	Fall 2021 - Spring 2024
Pupul Pradhan [†] (Utah Master → Evans & Sutherland)	Summer 2023
Novella Alvina [†] (Utah Master)	Fall 2024 - Summer 2025
John Chen (Utah Undergrad)	Spring 2025-Summer 2025

Workshop Organization Committees

Learning Effective Abstractions for Planning (LEAP) @ CoRL 2026 <i>Co-organizers: Naman Shah, Tom Silver, Utkarsh Mishra, David Paulius, Gregory Stein, Yiqing Xu, Amber Li, Franziska Herbert, Georgia Chalvatzaki</i>	<i>Austin, TX</i> Oct 2026
---	-------------------------------

Reviewing

Conference on Robot Learning (CoRL)	2023 - Present
Robotics: Science and Systems (RSS)	2024 - Present
Robotics and Automation Letters (RA-L)	2024 - Present
International Conference on Robotics and Automation (ICRA)	2023 - Present
International Conference on Intelligent Robots and Systems (IROS)	2024 - Present
International Conference on Learning Representations (ICLR)	2025 - Present
Neural Information Processing Systems (NeurIPS)	2026 - Present
IEEE Transactions on Robotics (T-RO)	2025 - Present
IEEE Transactions on Robot Learning (T-RL)	2026 - Present
The International Journal of Robotics Research (IJRR)	2026 - Present
IEEE Transactions on Artificial Intelligence	2024
IEEE Transactions on Instrumentation and Measurement	2024
IEEE Transactions on Industrial Electronics	2024
Workshop on Learning Effective Abstractions for Planning (LEAP @ CoRL)	2024 - Present
RSS Workshop on Out-of-Distribution Generalization in Robotics Towards Reliable Learning-Based Autonomy	2025

Skills

Computer Languages: C/C++, MATLAB, Python (TensorFlow, PyTorch), Java, VHDL

Software & Tools: IsaacGym, ROS, Gazebo, PyBullet, HTML, LaTeX

Outreach

Robotics lab tour co-organizer, University of Utah Bridge Program	2022, 2023
Introduction to mobile manipulator, Undergraduate AI Club at Princeton	2025
Introduction to robot programming for 7th-10th-grade students.	2026